

Week 5 Assignment: Module Five Activity Reflection

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Identifying Patterns

The visualization analysis of the College Scorecard Field of Study dataset revealed several clear patterns. Cleaning the data — which involved treating "PS" privacy-suppressed entries as missing values — reduced 229,188 records to 74 complete rows, all representing larger programs whose outcomes were publicly reportable. Ward's hierarchical clustering produced four well-separated groups of 13, 16, 23, and 22 records. The PCA scatterplot showed that two components captured approximately 73% of total variance, with the first component alone accounting for 53% — an unusually high concentration that suggests one dominant axis of variation runs through the data. Clusters formed visually distinct regions with minimal overlap. Both the feature importance heatmap and the trend chart identified earnings-count variables, such as EARN_COUNT_NE_3YR, as the strongest discriminators, and the scatterplot confirmed clusters separate primarily along these earnings dimensions.

Drawing Conclusions

Ultimately, these patterns demonstrate that post-graduation earnings distributions—rather than debt variables—primarily drive the natural clustering within large, fully reported programs. This finding is reinforced by several visualizations: a dendrogram revealing a clean four-way split at a high distance threshold, a PCA scatterplot confirming geometric separation, and a heatmap indicating that earnings features account for the highest cross-cluster variance.

How the Visualizations Validate the Findings

The visualizations validated these findings in ways raw numbers could not. The dendrogram's vertical axis made merge-strength immediately visible, while consistent viridis and plasma color scales linked cluster identity across views. PCA axes labeled with variance percentages quantified how much information the 2D projection preserved. Without the heatmap's color encoding, identifying the most discriminative features among 135 columns would have required scanning a large numeric table.

Summary of Visualization Choices

The analysis relied on six distinct chart types, each serving a specific purpose. While the dendrogram captured the hierarchical structure and the PCA scatterplot showed cluster geometry, an annotated bar chart illustrated cluster sizes. Meanwhile, a heatmap surfaced feature importance, a trend chart tracked a single variable across clusters, and a final scatterplot explored a two-feature relationship. Every graphic was designed for independent clarity, utilizing descriptive titles, fully labeled axes, explicit annotations, and a unified color scheme.

AI Usage Acknowledgment

Anthropic's Claude (claude.ai) was used to assist with Python code structure and errors. All analysis, interpretation, and editorial decisions are my own.